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BASIC MATHEMATICS

FORM FOUR

Certificate of Secondary Education Examination (CSEE) 2022 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Four
Year	2022
Exam Board	NECTA
Exam	Certificate of Secondary Education Examination (CSEE)
Type	Answer key and worked solutions
Questions	14 compulsory questions; 40 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F4-2022.pdf

Section A — Worked solutions

1(a). Find the percentage of multiples of 5 in $\{1,2,\dots,52\}$, to one decimal place.

Step 1: The multiples are 5,10,15,20,25,30,35,40,45,50: ten numbers.

Step 2: Percentage = $(10/52) \times 100 \approx 19.2308\%$, which rounds to 19.2%.

Final Answer: 19.2%.

1(b)(i). Arrange $1/2$, $2/9$, $3/8$, $1/12$ and $2/5$ in ascending order.

Step 1: Using denominator 360 gives $180/360, 80/360, 135/360, 30/360, 144/360$ respectively.

Step 2: Order the numerators 30,80,135,144,180 and restore the corresponding fractions.

Final Answer: $1/12 < 2/9 < 3/8 < 2/5 < 1/2$.

1(b)(ii). Simplify $(7 \times 10^4)/0.000035$ in standard form.

Step 1: $0.000035 = 3.5 \times 10^{-5}$.

Step 2: Divide coefficients and subtract exponents: $(7/3.5) \times 10^{[4 - (-5)]} = 2 \times 10^9$.

Final Answer: 2×10^9 .

2(a). Find x if $8^{(x-1)} = 16$.

Step 1: Rewrite $8 = 2^3$ and $16 = 2^4$.

Step 2: Equal bases give $3(x-1) = 4$, so $x = 1 + 4/3 = 7/3$.

Final Answer: $x = 7/3$.

2(b)(i). Simplify $\log_a \sqrt{a} + \log_a (a^2)$.

Step 1: Write $\sqrt{a} = a^{(1/2)}$. Then $\log_a a^{(1/2)} = 1/2$ and $\log_a a^2 = 2$.

Step 2: Add: $1/2 + 2 = 5/2$.

Final Answer: $5/2$, with $a > 0$ and $a \neq 1$.

2(b)(ii). Rationalise $(5 + \sqrt{2})/(\sqrt{6} - \sqrt{2})$.

Step 1: Multiply top and bottom by $\sqrt{6} + \sqrt{2}$. Denominator = $6 - 2 = 4$.

Step 2: Expand numerator: $(5 + \sqrt{2})(\sqrt{6} + \sqrt{2}) = 5\sqrt{6} + 5\sqrt{2} + \sqrt{12} + 2$; $\sqrt{12} = 2\sqrt{3}$.

Final Answer: $(5\sqrt{6} + 5\sqrt{2} + 2\sqrt{3} + 2)/4$.

3(a)(i). P contains multiples of 5 less than 35; Q contains odd numbers between 14 and 30. Find $P \cap Q$.

Step 1: $P = \{5, 10, 15, 20, 25, 30\}$; $Q = \{15, 17, 19, 21, 23, 25, 27, 29\}$.

Step 2: The intersection keeps only numbers present in both lists: 15 and 25.

Final Answer: $\{15, 25\}$.

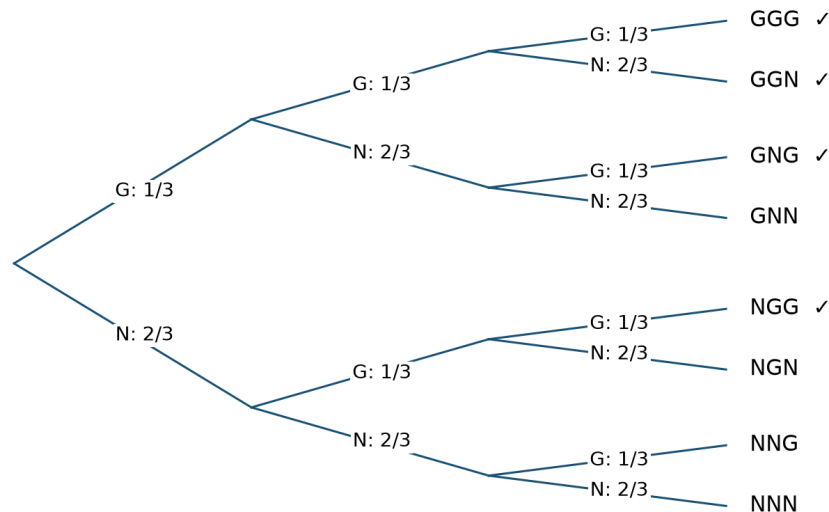
3(a)(ii). Of 50 farmers, 25 grow cashew nuts, 16 both cashews and maize, and 10 neither. Find maize only without a Venn diagram.

Step 1: At least one crop = $50 - 10 = 40$.

Step 2: The union consists of all cashew growers plus maize-only growers. Maize only = $40 - 25 = 15$.

Final Answer: 15 farmers.

3(b). Three seeds independently germinate with probability $\frac{1}{3}$ each. Use a tree diagram to find the probability at least two germinate.



Step 1: At each of three stages, branch to G with probability $\frac{1}{3}$ and N with probability $\frac{2}{3}$. Multiply probabilities along a path.

Step 2: Exactly two: GGN, GNG, NGG each has probability $(\frac{1}{3})^2(\frac{2}{3}) = \frac{2}{27}$; their total is $\frac{6}{27}$.

Step 3: All three: GGG has probability $(\frac{1}{3})^3 = \frac{1}{27}$. At least two = $\frac{6}{27} + \frac{1}{27} = \frac{7}{27}$.

Final Answer: $\frac{7}{27}$.

Study Tip: Multiplying branch probabilities assumes the seeds germinate independently, as intended in this probability model.

4(a). $\mathbf{a = (4,3)}$, $\mathbf{b = (-4,1)}$, $\mathbf{c = (2,5)}$. Compare lengths of $\mathbf{a + 2b}$ and $\mathbf{3a + c}$.

Step 1: $\mathbf{a + 2b = (4 - 8, 3 + 2) = (-4, 5)}$, length $\sqrt{16 + 25} = \sqrt{41}$.

Step 2: $\mathbf{3a + c = (12 + 2, 9 + 5) = (14, 14)}$, length $\sqrt{392} = 14\sqrt{2}$. Since $392 > 41$, the second vector is longer.

Final Answer: $\mathbf{3a + c}$ is longer: $\mathbf{14\sqrt{2} \approx 19.80}$, compared with $\mathbf{\sqrt{41} \approx 6.40}$.

4(b). Find the line through $\mathbf{(4,2)}$ perpendicular to $\mathbf{2x + 3y + 14 = 0}$.

Step 1: The given gradient is $-\frac{2}{3}$. Its perpendicular gradient is $\frac{3}{2}$.

Step 2: Use $y - 2 = (\frac{3}{2})(x - 4)$; multiply by 2 and rearrange.

Final Answer: $\mathbf{3x - 2y - 8 = 0}$ (or $\mathbf{y = \frac{3x}{2} - 4}$).

5(a). A triangle has sides 4,5,6 cm. A similar triangle's longest side is 18 cm. Find its other sides.

Step 1: Corresponding longest sides give scale factor $18/6 = 3$.

Step 2: Multiply the other lengths: $4 \times 3 = 12$ and $5 \times 3 = 15$ cm.

Final Answer: 12 cm and 15 cm.

5(b)(i). A regular hexagon inscribed in a circle has perimeter 72 cm. Find the circle radius.

Step 1: Each hexagon side = $72/6 = 12$ cm.

Step 2: Joining the centre to adjacent vertices creates a 60° isosceles triangle, hence an equilateral triangle. The radius equals the side, 12 cm.

Final Answer: 12 cm.

5(b)(ii). Triangle ABC has area 70 cm^2 , $AB = 14$ cm and $AC = 20$ cm. Find angle BAC.

Step 1: Use area = $(1/2)AB \times AC \times \sin A$: $70 = (1/2)(14)(20)\sin A$.

Step 2: $\sin A = 70/140 = 1/2$. For an interior triangle angle $0^\circ < A < 180^\circ$, both $A = 30^\circ$ and $A = 150^\circ$ satisfy this.

Final Answer: 30° or 150° ; the acute solution is 30° .

Study Tip: No diagram or acute-angle condition is supplied. Area and these two sides alone do not distinguish the two possible included angles.

6(a). Anna walks 24 km daily. Find her distance in two days in metres.

Step 1: Two-day distance = $2 \times 24 = 48$ km.

Step 2: Convert using $1 \text{ km} = 1000 \text{ m}$: $48 \times 1000 = 48,000 \text{ m}$.

Final Answer: 48,000 m.

6(b)(i). Cost price C is directly proportional to selling price s . One mattress sells for Tsh 20,000 and costs Tsh 18,000. Find the constant.

Step 1: Write $C = ks$.

Step 2: $18,000 = k \times 20,000$ gives $k = 18,000/20,000 = 0.9$.

Final Answer: $k = 0.9 = 9/10$.

6(b)(ii). Write the equation relating cost price and selling price.

Step 1: Direct variation has the form $C = ks$.

Step 2: Substitute the constant $k = 0.9$.

Final Answer: $C = 0.9s$ (equivalently $s = 10C/9$).

7(a). A damaged table costing Tsh 20,000 is sold at 15% loss. Find loss and selling price.

Step 1: Loss = $0.15 \times 20,000 = 3,000$.

Step 2: Selling price = $20,000 - 3,000 = 17,000$.

Final Answer: Loss = Tsh 3,000; selling price = Tsh 17,000.

7(b). Extract a trial balance from Mabala's cash account: capital 100,000; sales 43,000 and 47,000; purchases 80,000; telephone 28,000; cash balance 82,000.

Step 1: Cash receipts for capital and sales imply credit balances in their respective accounts; total sales = $43,000 + 47,000 = 90,000$.

Step 2: Purchases and telephone expenses are debit balances. Cash remaining, 82,000, is also a debit balance.

Step 3: Check debit total $80,000 + 28,000 + 82,000 = 190,000$; credit total $100,000 + 90,000 = 190,000$.

Final Answer: Debit balances: Purchases 80,000; Telephone expenses 28,000; Cash 82,000. Credit balances: Capital 100,000; Sales 90,000. Each total = Tsh 190,000.

Study Tip: Do not copy the cash-book sides directly into the trial balance: a cash receipt is a debit to Cash but a credit to its counterpart account.

8(a). The fifth and eleventh AP terms are 8 and -34 . Find the sum of the first ten.

Step 1: $a + 4d = 8$ and $a + 10d = -34$. Subtract: $6d = -42$, so $d = -7$. Then $a = 36$.

Step 2: $S_{10} = (10/2)[2a + 9d] = 5[72 - 63] = 45$.

Final Answer: 45.

8(b)(i). Tsh 100,000,000 is invested at 2% compounded annually. Find the amount after two years.

Step 1: $A = P(1 + r/100)^2 = 100,000,000(1.02)^2$.

Step 2: $1.02^2 = 1.0404$; multiply to obtain 104,040,000.

Final Answer: Tsh 104,040,000.

8(b)(ii). Find the interest earned after two years.

Step 1: Compound interest = accumulated amount – initial principal.

Step 2: $104,040,000 - 100,000,000 = 4,040,000$.

Final Answer: Tsh 4,040,000.

9(a)(i). Evaluate $\sin 690^\circ / \cos 690^\circ$ without tables.

Step 1: Subtract a full 360° : 690° has the same sine and cosine as 330° . The quotient is $\tan 330^\circ$.

Step 2: $330^\circ = 360^\circ - 30^\circ$, so $\tan 330^\circ = -\tan 30^\circ = -1/\sqrt{3} = -\sqrt{3}/3$.

Final Answer: $-\sqrt{3}/3$.

9(a)(ii). A 400 m by 300 m garden has seedlings along diagonal BD every 1.25 m, including both endpoints shown. How many seedlings?

Step 1: Diagonal length = $\sqrt{(400^2 + 300^2)} = 500$ m.

Step 2: Number of intervals = $500/1.25 = 400$. Including a seedling at each endpoint gives $400 + 1 = 401$.

Final Answer: 401 seedlings.

9(b). A car has angle of depression 30° from the top of a 50 m tower. Find its distance from the base in surd form.

Step 1: The angle of elevation from the car is also 30° . Let ground distance be d .

Step 2: $\tan 30^\circ = 50/d = 1/\sqrt{3}$, so $d = 50\sqrt{3}$.

Final Answer: $50\sqrt{3}$ m.

10(a). Express $2t^{-10} - 3t^{-5} + 1 = 0$ in terms of $x = 1/t^5$.

Step 1: $x = t^{-5}$, so $t^{-10} = (t^{-5})^2 = x^2$.

Step 2: Substitute these into the equation; t must be nonzero.

Final Answer: $2x^2 - 3x + 1 = 0$.

10(b). Use the quadratic formula to solve the equation in x .

Step 1: For $2x^2 - 3x + 1 = 0$, $a = 2$, $b = -3$, $c = 1$. Discriminant $b^2 - 4ac = 9 - 8 = 1$.

Step 2: $x = [3 \pm \sqrt{1}]/4 = (3 \pm 1)/4$, giving 1 and $1/2$.

Final Answer: $x = 1$ or $x = 1/2$.

Section Summary: Translate notation carefully: negative powers indicate reciprocals, vector length uses squares, and probability-tree paths multiply independent branch probabilities.

Section B — Worked solutions

11(a). An arc has length 22 cm and radius 63 cm. Find its central angle in degrees using $\pi = 22/7$.

Step 1: Circumference $= 2\pi r = 2 \times (22/7) \times 63 = 396$ cm.

Step 2: Angle $= (\text{arc/circumference}) \times 360^\circ = (22/396) \times 360^\circ = 20^\circ$.

Final Answer: 20° .

11(b). Prove x and y supplementary in the cyclic quadrilateral, with central angles a and b .

Step 1: Angle x at the circumference subtends the lower arc, so $x = a/2$. Angle y subtends the complementary upper arc, so $y = b/2$.

Step 2: Central angles around the point satisfy $a + b = 360^\circ$. Hence $x + y = (a + b)/2 = 180^\circ$.

Final Answer: $x + y = 180^\circ$, so x and y are supplementary.

11(c). Chords AB and CD intersect at E ; $AE = 8$ cm, $BE = 3$ cm, $CE = 4$ cm. Find DE .

Step 1: Intersecting-chord theorem: $AE \times BE = CE \times DE$.

Step 2: $8 \times 3 = 4 \times DE$; $DE = 24/4 = 6$ cm.

Final Answer: 6 cm.

12(a). A bus travels from $A(3^\circ S, 39^\circ E)$ to $B(12^\circ S, 39^\circ E)$ at 40 km/h. $R = 6400$ km, $\pi = 3.14$. Find travel time.

Step 1: The same-meridian angle is $12 - 3 = 9^\circ$. Distance = $(9/360) \times 2 \times 3.14 \times 6400 = 1004.8$ km.

Step 2: Time = distance/speed = $1004.8/40 = 25.12$ hours.

Final Answer: 25.12 hours (25 hours 7.2 minutes).

12(b)(i). Cuboid $UVXY$ has $UV = 4.2$ cm, $VX = 2$ cm and $XR = 2.5$ cm. Find VR and UR to one decimal place.

Step 1: VR is the side-face diagonal: $VR^2 = VX^2 + XR^2 = 2^2 + 2.5^2 = 10.25$, so $VR \approx 3.2016$ cm.

Step 2: UR is the space diagonal: $UR^2 = UV^2 + VX^2 + XR^2 = 17.64 + 4 + 6.25 = 27.89$, so $UR \approx 5.2811$ cm.

Step 3: Round each length to one decimal place.

Final Answer: $VR = 3.2$ cm; $UR = 5.3$ cm.

12(b)(ii). Find the angle between UR and base UVXY.

Step 1: The projection of R onto the base is X. $UX = \sqrt{(4.2^2 + 2^2)} = \sqrt{21.64} \approx 4.6519$ cm.

Step 2: $\tan \angle RUX = XR/UX = 2.5/\sqrt{21.64}$; the angle is approximately 28.25° .

Final Answer: $\angle RUX \approx 28.3^\circ$.

Study Tip: Use unrounded lengths when finding the angle, rather than the rounded 5.3 cm diagonal.

13(a). Solve $\begin{bmatrix} x & 4 \\ 4 & y \end{bmatrix} \begin{bmatrix} -5 & -7 \\ 2 & z \end{bmatrix} = \begin{bmatrix} 38 & 46 \\ -10 & w \end{bmatrix}$.

Step 1: Multiply matrices entry by entry: $-5x + 8 = 38$, $-7x + 4z = 46$, $-20 + 2y = -10$, $-28 + yz = w$.

Step 2: First $x = -6$ and $y = 5$. Then $42 + 4z = 46$ gives $z = 1$. Finally $w = -28 + 5 = -23$.

Final Answer: $x = -6$; $y = 5$; $z = 1$; $w = -23$.

13(b). Use matrices to reflect $(3, -2)$ in $y = -x$, then in $x = 0$.

Step 1: The first reflection matrix is $S = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$; the second is $Y = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.

Step 2: Apply in the given order: $YS = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Multiply YS by $\begin{bmatrix} 3 \\ -2 \end{bmatrix}$ to get $\begin{bmatrix} -2 \\ -3 \end{bmatrix}$.

Step 3: Intermediate check: first image $(2, -3)$, then y-axis reflection $(-2, -3)$.

Final Answer: $(-2, -3)$.

13(c). A translation takes $(5, 5)$ to $(-7, -7)$, and (x, y) to $(-4, -4)$. Find x and y .

Step 1: Translation vector $= (-7 - 5, -7 - 5) = (-12, -12)$.

Step 2: $(x - 12, y - 12) = (-4, -4)$, so $x = y = 8$.

Final Answer: $x = 8$; $y = 8$.

14(a)(i). For $f(x) = 1/(x - 2)$, give domain and range.

Step 1: The denominator cannot be zero, so $x \neq 2$.

Step 2: The numerator is 1, so output 0 is impossible. Any nonzero output y is attained at $x = 2 + 1/y$.

Final Answer: Domain: $\mathbb{R}$ excluding 2. Range: $\mathbb{R}$ excluding 0.

14(a)(ii). Find $f^{-1}(1/3)$.

Step 1: Set $y = 1/(x - 2)$. Then $x - 2 = 1/y$, so $f^{-1}(y) = 2 + 1/y$.

Step 2: Substitute $y = 1/3$: $2 + 3 = 5$. Check $f(5) = 1/3$.

Final Answer: 5.

14(b)(i). Black shirts cost Tsh 24,000, white Tsh 30,000; buy at most 5 black and spend at most Tsh 180,000. Maximise total shirts.

Step 1: Let nonnegative integers x, y count black and white shirts. Constraints: $x \leq 5$ and $24,000x + 30,000y \leq 180,000$, or $4x + 5y \leq 30$.

Step 2: For each $x = 0, 1, 2, 3, 4, 5$, the greatest possible y is 6, 5, 4, 3, 2, 2; total shirts are 6, 6, 6, 6, 6, 7.

Step 3: Best is (5, 2), costing $5 \times 24,000 + 2 \times 30,000 = 180,000$.

Final Answer: Buy 5 black shirts and 2 white shirts.

14(b)(ii). Find the greatest total number of shirts.

Step 1: The optimum combination is 5 black and 2 white.

Step 2: Total = $5 + 2 = 7$; eight shirts are impossible because even the cheapest allowed mix costs at least $5(24,000) + 3(30,000) = 210,000$.

Final Answer: 7 shirts.

Section Summary: Circle theorems relate central and inscribed angles or products of chord segments. In matrix compositions, the transformation applied first is on the right. Enforce whole-number constraints when counting shirts.